Date: September 29, 2026

To,
The Editor,
2026 International Conference on Intelligent Systems & Data-driven Computing (ISDDC)

Subject: Paper ID 520
Title: Multimodal Neonatal Pain Classification Using EEG, ECG, and Respiration Signals with Subject-Level Validation
Track Name: Special Track 2 [Advanced Computational Intelligence and Smart Digital Technologies (Optimization, AI & ML, Data Analytics, Cloud Intelligence, IoT, Cybersecurity, and Smart Applications)]
Authors: Arpit Pandey, Alka Verma, Vimal Mishra, and Yogendra Kumar Prajapati

Respected Editor,

This has a reference to your email dated September 12th, 2026 forwarding the comments of the reviewers on the above manuscript. Thank you very much for giving us an opportunity to revise the paper in the light of the comments of the reviewers.

The comments of the editors and reviewers were of immense help in improving the technical content of the paper. Please find the revised version of the manuscript uploaded along with the detailed response to the comments of the reviewers.

With best regards,

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Response to Reviewers' Comments

Manuscript ID: 520
Title: Multimodal Neonatal Pain Classification Using EEG, ECG, and Respiration Signals with Subject-Level Validation

Response to Reviewer #1

Recommendation: Accept with Minor Revisions

The authors are highly thankful to the learned reviewer for their valuable suggestions, comments, and modifications towards increasing the strength and readability of the manuscript. The manuscript has been accordingly revised. The suggestions, comments, and modifications are addressed below.

Comment #1: "The reported AUCs (0.849 ± 0.285; 0.929 ± 0.175) show substantial variability, yet no confidence intervals or statistical comparisons support the claims of competitive performance. F1-scores are also low and highly variable (0.475 ± 0.450; 0.452 ± 0.438), despite strong AUCs, requiring analysis of threshold selection or class imbalance. The small PIPP cohort (n=28) also limits LOSO generalisability."

Response: We sincerely thank the reviewer for this insightful observation. We have comprehensively addressed every part of this comment in the revised manuscript as follows:

1. **Addition of 95% Confidence Intervals:** We have updated Table IV in the manuscript to include 95% Confidence Intervals (CIs) for the mean subject-level AUCs. The Broad Condition-based cohort now reports an AUC of 0.849 ± 0.287 with a 95% CI of [0.770, 0.928], and the PIPP-confirmed cohort reports 0.929 ± 0.178 with a 95% CI of [0.859, 0.998]. This directly quantifies the between-subject variability in LOSO evaluation.
2. **Clarification of "Mean LOSO AUC":** We have added a clear explanation in the *Methods* section stating that the reported primary AUC is the arithmetic mean of the subject-level AUCs obtained from each held-out subject in the LOSO cross-validation. The SD and 95% CI are calculated across these subject-level AUCs, representing between-subject variability rather than uncertainty from random splits.
3. **Distinction Between Mean Subject-Level AUC and Pooled AUC:** We have clarified in the *Results* section that the ROC curves in Fig. 5 are constructed from pooled out-of-fold (OOF) predictions across LOSO folds (yielding pooled AUCs of 0.806 and 0.810), whereas the AUC values in Table IV are arithmetic means of subject-level AUCs. These are complementary metrics answering different questions, and we have explicitly labeled them to prevent confusion.

4. **Explanation of AUC vs. F1 Discrepancy:** We have added a dedicated paragraph in the *Discussion* explaining that AUC evaluates continuous ranking across all thresholds, whereas F1 depends on a specific decision threshold (default ≈ 0.5). Furthermore, the high variability in subject-level F1 is partly attributable to the very small number of test samples available for individual subjects under LOSO evaluation (often 2–3 samples), making F1 highly sensitive to individual predictions.

5. **Class Imbalance and Threshold Sensitivity Analysis:** We have added a fixed-threshold sensitivity analysis in the *Results* section, demonstrating that F1 is sensitive to the operating threshold (e.g., maximum observed F1 ≈ 0.672 at threshold 0.40 for the broad protocol). We explicitly state that leakage-free threshold optimization and probability calibration within the LOSO framework are reserved for future work.

6. **Strengthened Limitations Regarding Cohort Size:** We have explicitly stated in the *Limitations* section that the PIPP-confirmed analysis is available for only 28 subjects, which limits generalisability to the available cohort. We have also reinforced that external validation on larger, independent, multi-centre neonatal datasets is a critical next step.

7. **Softened Claims:** We have carefully reviewed the manuscript and softened claims of "competitive performance" and "improvement" to accurately reflect that the observed differences are between evaluation protocols with different label definitions and cohort sizes, rather than a direct statistical superiority.

Response to Reviewer #3

Recommendation: Accept with Minor Revisions

The authors are highly thankful to the learned reviewer for their valuable suggestions, comments, and modifications towards increasing the strength and readability of the manuscript. The manuscript has been accordingly revised. The suggestions, comments, and modifications are addressed below.

Comment #1 (External Validation): Clarify LOSO $\neq$ external validation. Strengthen limitations. State current generalisability boundary. Add independent/multi-centre validation as future work.

Response: We completely agree with the reviewer. We have explicitly clarified in the *Discussion* and *Limitations* sections that LOSO cross-validation provides subject-independent evaluation *within* the available UCL cohort, but it does not establish external clinical generalisability to the broader neonatal population. We have strengthened the limitations to state that the

PIPP-confirmed analysis is limited to 28 evaluable subjects, and we have explicitly added that external validation on independent, larger, multi-centre neonatal cohorts is a mandatory future step to confirm broader clinical applicability.

Comment #2 (Model Interpretability): Current SHAP notebook is incomplete. Do not invent results. Add interpretability as future work/limitation. Actual SHAP analysis can be developed later.

Response: We thank the reviewer for this prudent guidance. We have not included any preliminary or incomplete SHAP results in the manuscript. Instead, we have explicitly added "detailed model interpretability using explainable machine learning methods such as SHAP" to the *Future Work* and *Limitations* sections. We agree that a comprehensive SHAP-based explainability analysis (including global beeswarm plots and per-modality importance) is a critical next step for clinical trust and will be the primary focus of our planned journal extension.

Comment #3 (Computational Analysis): 69 features. 200 XGBoost estimators. Max depth 4. Learning rate 0.05. Fold-specific class weighting. Mean LOSO training time: 0.235 s. Median LOSO training time: 0.213 s. Mean LOSO inference time: 0.00225 s. Median inference time: 0.00204 s. 200 trees. Hardware/software details to be recorded from Cell 1. Add one concise computational-efficiency paragraph. No large benchmark.

Response: We appreciate the reviewer's suggestion to quantify computational efficiency. We have added a concise paragraph in the *Classification* subsection detailing the computational requirements. The revised text now states: *"The computational requirements recorded during the LOSO analysis showed high efficiency. Across evaluated folds, mean training time was 0.235s per held-out subject (median: 0.213s), while mean inference time was 0.00225s (median: 0.00204s). All experiments were conducted on a Windows system with a 12-core Intel processor using Python 3.10, scikit-learn 1.7.2, and XGBoost 3.2.0, utilizing 69 features and 200 estimators with a max depth of 4 and learning rate of 0.05, alongside fold-specific class weighting."* This provides the exact transparency requested without overextending into a large benchmark.

Response to Reviewer #4

Recommendation: Accept with Minor Revisions

The authors are highly thankful to the learned reviewer for their valuable suggestions, comments, and modifications towards increasing the strength and readability of the manuscript. The manuscript has been accordingly revised. The suggestions, comments, and modifications are addressed below.

Comment #1: "Improve the citations and add some recent ones too."

Response: We sincerely thank the reviewer for this constructive suggestion. We have carefully revised the Related Work section to improve citation placement and have incorporated recent, highly relevant literature on neonatal pain assessment using physiological signals and multimodal intelligent methods. Specifically, the following updates have been made to the manuscript:

1. **Addition of Recent EEG-Based Neonatal Pain Study:** We have added the 2024 study by Shafiee and Daliri, titled "Decoding of pain during heel lancing in human neonates with EEG signal and machine learning approach" (*Scientific Reports*, vol. 14, 2024), which focuses on decoding pain during heel lancing in human neonates using EEG signals and machine learning. This citation has been integrated into the discussion of EEG-based machine learning approaches for neonatal pain.

2. **Addition of Recent Multimodal Assessment Review:** We have also included the 2026 systematic review by Zhou *et al.*, titled "Application of Intelligent Neonatal Pain Assessment Tools Based on Multimodal Data Fusion: A Systematic Review" (*Journal of Pain Research*, vol. 19, 2026). This reference has been incorporated to highlight the growing use of multimodal intelligent assessment methods in the field.

3. **Improved Citation Placement and Contextual Comparison:** The revised Related Work section now places citations directly alongside the corresponding claims. The discussion now seamlessly includes Hamwi *et al.* (2025), Roué *et al.* (2025), Shafiee and Daliri (2024), and Zhou *et al.* (2026). Furthermore, we have explicitly clarified in the text that direct numerical comparison between these studies is limited due to differences in datasets, label definitions, feature sets, and validation procedures.

We believe these additions significantly strengthen the manuscript's literature foundation and properly position our work within the current state-of-the-art.
